

Diagonal Tree Tensor Networks

Hierarchical Feature Learning through Shared Factors and Residual Paths

Definitions, exact gradient flow, causal tests, and feature extraction

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Abstract

A diagonal tree tensor network (TTN) combines matching channels by coordinatewise multiplication. This makes it a natural candidate for a tractable nonlinear extension of diagonal deep linear networks (DLNs), but the architectural analogy does not by itself imply a new learning mechanism. With fixed orthogonal leaf features, a homogeneous, root-readout diagonal TTN collapses to independent scalar product factorizations: its amplitude dynamics are exactly those of diagonal DLNs with the same number of factors. Nested staircase readouts change this conclusion by making lower-order and higher-order coefficients share parameters. We derive their gradient flow, hard gating, and coupled early-time acquisition clocks. Channelwise residual streams instead provide direct lower-order learning routes and additional learned compositional routes. We derive an exact three-leaf acceleration example, generalize its intervention calculation to arbitrary binary trees, and give a linear-cost population-moment recursion. The resulting soft acceleration is observable-dependent and need not imply a one-way or bottom-up training order. Finally, we distinguish amplitude acquisition from discovery of unknown directions, describe function-space and tangent-feature extraction, and position these models relative to the project's literature review. Proven identities, source results, and proposed extensions are separated throughout.

Central conclusion. Diagonality separates channels; it does not create a hierarchy. The tractable mechanism is *parameter sharing across distinct, statistically distinguishable feature coefficients*. Staircase readouts make that sharing prerequisite-like. Residual paths can make it facilitating rather than strictly necessary.

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1 Scope, sources, and the question being answered

The question is whether we can explain, with explicit training equations, how acquiring one representation makes another easier to learn. The proposed sequence of models is

$$\text{homogeneous diagonal TTN} \longrightarrow \text{staircase or residual channelwise TTN}.$$

The first is a control model. The latter two introduce several output features whose coefficients share trainable factors. Throughout, “TTN” refers to a bilinear tree with distinct input blocks at its leaves, not to a transformer or to a sequential network evaluated on repeated copies of one input.

The main sources are the teacher–student setup draft [P1], the residual-TTN research note [P2], the literature review [P3], and the attached feature-ODE paper [P4]. The definitions and proofs below expand these sources. In particular, the staircase onset calculation and the three-leaf residual example are already in the project notes; they are not presented as new literature results. The general residual support expansion and its arbitrary-tree intervention theorem are derivations in this document, not claims of research priority. Selected external papers from the review were checked against their primary sources as of the date on the cover.

1.1 Three qualifications to the earlier discussion

First, “learning one feature speeds up another” is too weak to exclude DLNs unless we specify what the features are. Two factors of one scalar product can facilitate each other in a DLN as well as in a TTN. Distinct, independently supervised singular modes are a different kind of observable.

Second, a residual TTN can have *soft causal acceleration without a directed acquisition order*. An upper parameter can learn before a lower one, and the coupling can be reciprocal. The residual project note explicitly makes this distinction [P2, Sections 1.1 and 6.3].

Third, a channelwise residual TTN is not a homogeneous TTN with all augmented cores superdiagonal. Its bilinear part is diagonal, but the residual stream needs additional routing entries. Similarly, staircase readouts modify the observation architecture; they are not obtained merely by choosing a different target for a homogeneous root-only student.

1.2 What the basic calculations do and do not model

The core analysis fixes orthogonal primitive directions and learns scalar gains and composition strengths. It therefore isolates *acquisition, amplification, and reuse in a prescribed feature family*. It does not establish that a generic randomly initialized network discovers unknown primitive directions, an unknown tree, or arbitrary combinations of different channels. Section 8 gives exact directional extensions and identifies where the scalar closure ceases to apply.

All gradient flows below are Euclidean population gradient flow with unit learning rates and no weight decay, unless stated otherwise. Freezing gates, changing their parameterization, or optimizing factorized versions of the same effective core changes the dynamics. A tensor identity alone is not an equivalence between learning algorithms.

2 Definition of a diagonal TTN and the statistical setting

2.1 Trees, cores, and matching channels

Let $\mathcal{T}$ be a rooted full binary tree with $N \geq 2$ leaves. A leaf i receives $x_i \in \mathbb{R}^{d_i}$. At an internal node v , write v_L, v_R for its children and S_v for the set of leaves below it. A generic width- r bilinear core computes

$$h_{v,k} = \sum_{p,q=1}^r B_{kpq}^{(v)} h_{v_L,p} h_{v_R,q}. \quad (2.1)$$

Definition 2.1 (Channel-diagonal TTN). Fix compatible channel coordinates $k = 1, \dots, r$ on the bonds. A *channel-diagonal* internal core is

$$\boxed{B_{kpq}^{(v)} = g_{v,k} \delta_{kp} \delta_{kq}}, \quad h_{v,k} = g_{v,k} h_{v_L,k} h_{v_R,k}. \quad (2.2)$$

The leaf maps have the form

$$h_{i,k} = a_{i,k} z_{i,k}, \quad z_{i,k} = u_{i,k}^\top x_i. \quad (2.3)$$

A scalar root can be written

$$f_\theta(x) = \sum_{k=1}^r g_{o,k} h_{o_L,k} h_{o_R,k}. \quad (2.4)$$

Thus the root's two-index core is diagonal. Equivalently, one may compute r root channels and apply a fixed summation readout.

A common channel label is transported through the whole tree: $(k, k) \mapsto k$. The vectors $u_{i,k}$ may differ between leaves; matching labels do not mean the two children represent the same function. What is excluded is a learned mixing rule $(p, q) \mapsto k$ with arbitrary p, q, k .

The algebraic definition does not require the $u_{i,k}$ to be orthogonal. Orthogonality and whether these directions are fixed or learned are additional assumptions. In particular, *diagonal cores alone do not guarantee independent learning channels*.

2.2 The fixed, orthogonal regime used for exact decoupling

Assumption 2.2 (Independent blocks and fixed orthogonal features). The blocks $x_1, \dots, x_N$ are independent, centered, and satisfy $\mathbb{E}[x_i x_i^\top] = I$. For every leaf, the fixed vectors $u_{i,1}, \dots, u_{i,r}$ are orthonormal. Student and teacher use the same vectors. Only the specified scalar parameters are trained.

Gaussianity is not required. Nor is independence between coordinates within the same block: the calculations only use the stated block independence and second moments. For nonempty $S \subseteq [N]$, define

$$\Psi_{S,k}(x) = \prod_{i \in S} z_{i,k}. \quad (2.5)$$

Then

$$\mathbb{E}[\Psi_{S,k}] = 0, \quad \boxed{\mathbb{E}[\Psi_{S,k} \Psi_{T,\ell}] = \delta_{ST} \delta_{k\ell}}. \quad (2.6)$$

If $S \neq T$, a block in the symmetric difference appears once and centering kills the expectation. If $S = T$, independence gives a product of within-block inner products, each equal to $\delta_{k\ell}$.

Consequently, if

$$f_\theta = \sum_{S,k} c_{S,k}(\theta) \Psi_{S,k}, \quad f_\star = \sum_{S,k} c_{S,k}^\star \Psi_{S,k}, \quad (2.7)$$

then the population loss is exactly

$$\mathcal{L}(\theta) = \frac{1}{2} \mathbb{E}[(f_\theta - f_\star)^2] = \frac{1}{2} \sum_{S,k} (c_{S,k}(\theta) - c_{S,k}^\star)^2. \quad (2.8)$$

This is one scalar regression target, not a separate training task for every term. Orthogonality makes the loss additive; shared parameters can nevertheless couple its terms. This is the independent-block tensor-regression reduction used in the project drafts [P1, Sections 3–4][P2, Section 5].

2.3 Why diagonal edge SVDs are not the definition

Cutting an edge partitions the physical leaves into S and S^c . Every corresponding matricization admits an SVD,

$$T^{\langle S|S^c \rangle} = U_S \Sigma_S V_S^\top. \quad (2.9)$$

Existence of such an SVD is generic, not a restriction to diagonal networks. Even a canonical TTN can have dense transfer cores joining child singular modes to parent singular modes. The diagonal restriction is the stronger *simultaneous, compatible no-mixing condition* in equation (2.2). Hierarchical SVD is a tool for analyzing general TTNs, not a justification for replacing their cores by superdiagonal ones [2][P2, Section 7].

2.4 Collapse to a CP tensor

Let $\mathcal{P}$ index the scalar factors actually trained in a homogeneous channel: leaf gains, internal gains, and the root gain as appropriate. Take any remaining fixed gains to be one. Repeated substitution gives

$$f_\theta(x) = \sum_{k=1}^r s_k(\theta) \Psi_{[N],k}(x), \quad s_k = \prod_{p \in \mathcal{P}} \theta_{p,k}. \quad (2.10)$$

Its coefficient tensor is

$$T_\theta = \sum_{k=1}^r s_k u_{1,k} \otimes \cdots \otimes u_{N,k}. \quad (2.11)$$

This is a CP decomposition: a sum of rank-one tensor products. Under proposition 2.2 it is orthogonally decomposable. Without that assumption it is simply a CP representation of rank at most r .

Across every nontrivial cut, the vectors $\bigotimes_{i \in S} u_{i,k}$ and $\bigotimes_{i \notin S} u_{i,k}$ are orthonormal. Hence every cut has the same nonzero singular-value multiset

$$\{\sigma_{S,k}\}_k = \{|s_k|\}_k. \quad (2.12)$$

Signs are absorbed into singular vectors; degenerate values determine subspaces rather than unique vectors. This equality of cut spectra is a major restriction. Increasing r adds channels, but does not by itself create independent rank-acquisition processes at different levels of the tree.

3 What counts as hierarchical feature learning?

3.1 A causal, observable-based criterion

A feature can mean a primitive direction, a nonlinear function of a subtree, an output coefficient, a singular subspace, or a tangent feature seen by a parameter. These are not interchangeable. Fix the data law, parameterization, optimizer, observable $q_B(\theta)$, target value $q_B^* > 0$, and a proposed earlier computation A . The choice of A and B should precede inspecting training curves.

Definition 3.1 (Counterfactual acquisition time). For a fixed threshold $\eta \in (0, 1)$ and a specified intervention $\mathcal{I}$, let

$$\tau_B(\eta \mid \mathcal{I}) = \inf\{t \geq 0 : q_B(\theta_{\mathcal{I}}(t)) \geq \eta q_B^*\}. \quad (3.1)$$

The infimum is $+\infty$ if no crossing occurs. Compare a learned state of A with a blocked state while keeping the same initialization of the remaining free variables and the same learning rates.

The weak acceleration condition is

$$\tau_B(\eta \mid A \text{ learned}) < \tau_B(\eta \mid A \text{ blocked}). \quad (3.2)$$

It is *hard* when the right-hand side is infinite, and *soft* when both times are finite. This follows the intervention-based definition in the residual note [P2, Section 1.2]. A local diagnostic compares the target-directed velocity at matched states; it is not a substitute for a hitting-time comparison, because the sign can change later in training.

For a stronger claim of *hierarchical* acquisition, the facilitating computation should sit below and be reused by the downstream computation. Evidence of a directed order additionally requires asymmetry: lower computations can acquire signal without later ones, whereas later acquisition depends on or benefits from the earlier state. The project setup uses stronger hard-gating, coupled-onset, and intervention conditions for its staircase model [P1, Section 5]; soft models need not satisfy that complete definition.

3.2 Three distinctions that prevent false positives

Staging versus causation. Two independent modes can be learned at different times because their target strengths differ. This is not evidence that the first changes the second’s learning rate.

Mutual facilitation versus directed hierarchy. Factors of one product can accelerate each other symmetrically. The weak condition in equation (3.2) can detect that, including in a DLN. A bottom-up feature-learning interpretation needs more structure than the existence of positive cross-derivatives.

Amplification versus direction discovery. A coefficient becoming nonzero is a meaningful acquisition event, but it does not demonstrate that an unknown direction was discovered. If all directions are supplied to the student, that part of the representation-learning problem has been removed.

3.3 Gauge and intervention controls

In a product parameterization, shrinking one factor can be offset by growing another. Thus raw hidden amplitudes are generally not gauge-invariant feature strengths. An intervention on a scalar gate is a legitimate experiment in a fixed architecture, but it should not be called a coordinate-free corruption of a learned direction. For directional tests, compare aligned and orthogonal vectors of the same norm, or intervene on a function-space subspace. The setup draft gives explicit examples where magnitude clamping permits compensation while directional misalignment limits a composite output [P1, Sections 5.1–5.3].

4 Homogeneous diagonal TTNs: nonlinear functions, DLN-like dynamics

4.1 Exact gradient flow and factor balance

For a teacher $f_\star = \sum_k s_k^\star \Psi_{[N],k}$, equations (2.8) and (2.10) give

$$\mathcal{L} = \frac{1}{2} \sum_k (s_k - s_k^\star)^2. \quad (4.1)$$

Differentiation yields, without dividing by a possibly zero parameter,

$$\dot{\theta}_{p,k} = (s_k^\star - s_k) \prod_{q \in \mathcal{P} \setminus \{p\}} \theta_{q,k}. \quad (4.2)$$

Different k are independent. Within one k , every factor enters the same product and gates the others symmetrically. The coefficient dynamics is

$$\dot{s}_k = K_k(\theta)(s_k^\star - s_k), \quad K_k = \sum_{p \in \mathcal{P}} \left(\prod_{q \neq p} \theta_{q,k} \right)^2. \quad (4.3)$$

For any two factors in a channel,

$$\frac{d}{dt}(\theta_{p,k}^2 - \theta_{q,k}^2) = 0, \quad (4.4)$$

since each squared factor has derivative $2s_k(s_k^\star - s_k)$.

Proposition 4.1 (Optimization collapse). *Under proposition 2.2, with the parameterization in equation (2.10), homogeneous diagonal TTN training is exactly the scalar amplitude training of a diagonal DLN with $m = |\mathcal{P}|$ factors per mode and the same target strengths and initialization.*

Proof. A diagonal DLN has effective mode coefficient $\prod_{p=1}^m w_{p,k}$. Its orthogonal-mode square loss is equation (4.1); differentiation gives the same vector field equation (4.2). The tree topology no longer appears. $\square$

This is an equality of these restricted optimization problems, not an equivalence between the full function classes or generic training trajectories. It is consistent with the aligned-mode analysis of Saxe, McClelland, and Ganguli [1].

4.2 An exact small-initialization comparison

If all m factors of a channel start equally and positively, balance and uniqueness preserve $\theta_{p,k} = a_k$. For $s_k^\star > 0$,

$$\dot{a}_k = (s_k^\star - a_k^m) a_k^{m-1}, \quad \dot{s}_k = m(s_k^\star - s_k) s_k^{2-2/m}. \quad (4.5)$$

Starting at $a_k(0) = \varepsilon$, the time to a fixed fractional coefficient $s_k = \eta s_k^\star$ is exactly

$$\tau_k = \int_{\varepsilon}^{(\eta s_k^\star)^{1/m}} \frac{da}{(s_k^\star - a^m) a^{m-1}}. \quad (4.6)$$

For $m > 2$,

$$\tau_k \sim \frac{\varepsilon^{-(m-2)}}{(m-2)s_k^\star} \quad (\varepsilon \downarrow 0), \quad (4.7)$$

while $m = 2$ gives a logarithmic divergence. With $m \geq 2$, initializing all factors exactly at zero is stationary. Initializing only one factor at zero need not be stationary.

The count m is not the graph depth. With trainable leaf and internal scalar gains in a full N -leaf tree, $m = 2N - 1$; with fixed leaves and only internal gains, $m = N - 1$. A comparison to a DLN must match the factor count and parameter-space metric, rather than simply giving both architectures the same number of levels.

4.3 Does it have the requested acceleration property?

Between independent output modes s_k and s_ℓ : no. An intervention on channel k leaves the entire differential equation for channel $\ell \neq k$ unchanged. The channel cut features in equation (2.12) are fixed in direction, up to sign, ordering, and degenerate-subspace conventions, and share the same amplitudes at every cut.

Between factors within one channel: weakly, yes. With positive other factors and $p \neq q$,

$$\frac{\partial \dot{\theta}_{p,k}}{\partial \theta_{q,k}} = \left(\prod_{u \notin \{p,q\}} \theta_{u,k} \right) (s_k^\star - 2s_k), \quad (4.8)$$

which is positive early in learning. But this is the same mutual product facilitation as in the matched DLN, not a directed discovery of distinct nonlinear representations. This is why the weak definition must not be used to assert that all DLNs have no feature interactions.

As a model of bottom-up nonlinear feature acquisition: not in this fixed-basis regime. A subtree computes a higher-degree product, but its normalized function is already specified. Balanced factors grow together, and the root-only loss does not separately identify or reward intermediate composites. Rebracketing the products changes the computational tree without changing the amplitude equations.

4.4 What nevertheless differs from a DLN

The output is nonlinear in the raw input blocks: $\Psi_{[N],k}$ has degree N , whereas a DLN remains linear in its input. The hidden computations also contain nonlinear products. Thus this TTN represents nonlinear targets, but in the fixed orthogonal restriction its *amplitude learning* is equivalent to a diagonal DLN acting on an already specified nonlinear feature dictionary. Learning the leaf directions or allowing non-diagonal composition changes that restricted conclusion.

5 Staircase diagonal TTNs: hard gating and coupled clocks

5.1 Architecture and teacher–student objective

Use a comb-shaped tree and a fixed readout from every prefix. For $j = 1, \dots, N$ and channel k , define

$$H_{0,k} = 1, \quad H_{j,k} = a_{j,k} z_{j,k} H_{j-1,k}, \quad p_{j,k} = \prod_{i \leq j} a_{i,k}. \quad (5.1)$$

The student and teacher are

$$f_a(x) = \sum_{k=1}^r \sum_{j=1}^N \sqrt{\lambda_{j,k}} p_{j,k} \Psi_{[j],k}(x), \quad (5.2)$$

$$f_\star(x) = \sum_{k=1}^r \sum_{j=1}^N \sqrt{\lambda_{j,k}} p_{j,k}^\star \Psi_{[j],k}(x), \quad p_{j,k}^\star = \prod_{i \leq j} a_{i,k}^\star. \quad (5.3)$$

All $\lambda_{j,k} > 0$ are fixed. The $a_{j,k}$ are the only trained scalars; training an additional gate at every multiplication would be a different parameterization. Under [proposition 2.2](#),

$$\mathcal{L}_{\text{stair}}(a) = \frac{1}{2} \sum_{k,j} \lambda_{j,k} (p_{j,k} - p_{j,k}^\star)^2. \quad (5.4)$$

The same $a_{i,k}$ participates in every later prefix, so the loss is orthogonal in *output features*, but not separable in *parameters*. The hierarchy is in this nested sharing. There is still only one scalar teacher label.

For the cleanest gating and onset results, take $a_{i,k}^\star = 1$. This is a chosen teacher family, not an assertion that an arbitrary rescaling of teacher parameters preserves Euclidean training time.

Source and status. This is the fixed-direction, multi-channel version of the prefix model in the setup draft [P1, Sections 4–5]. The internal channelwise multiplication is diagonal, but the added prefix readouts put the model outside the strictly homogeneous root-only class of Section 2.

5.2 Gradient flow: derivation and gating

Suppress k . Let $C_{m,j} = \prod_{i \leq m, i \neq j} a_i$. Then

$$\frac{\partial p_m}{\partial a_j} = \begin{cases} C_{m,j}, & m \geq j, \\ 0, & m < j, \end{cases} \quad \dot{a}_j = \sum_{m \geq j} \lambda_m (p_m^\star - p_m) C_{m,j}. \quad (5.5)$$

For unit teacher factors, separate the signal and self-interaction terms:

$$D_j(a) = \sum_{m \geq j} \lambda_m C_{m,j}^2, \quad (5.6)$$

$$\dot{a}_j = -D_j(a) a_j + \left(\prod_{i < j} a_i \right) \left[\lambda_j + \sum_{m > j} \lambda_m \prod_{j < i \leq m} a_i \right]. \quad (5.7)$$

The first feature has an ungated λ_1 drive. Every target-directed contribution to a_j includes every earlier factor. From nonnegative initial conditions the positive orthant is invariant: at $a_j = 0$ its derivative is nonnegative.

For scalar amplitudes, clamping any earlier a_i to zero also makes $D_j = 0$, so a_j cannot move. In the directional extension, zero *alignment* can coexist with nonzero norm; then the target-directed signal vanishes but a shrinkage term may remain. The distinction matters when interpreting “no gradient.”

5.3 A hard acceleration test in the two-stage model

Take two features and unit teacher factors:

$$\mathcal{L}(a, b) = \frac{\lambda_1}{2} (a - 1)^2 + \frac{\lambda_2}{2} (ab - 1)^2, \quad (5.8)$$

so

$$\dot{a} = \lambda_1(1 - a) + \lambda_2 b(1 - ab), \quad \dot{b} = \lambda_2 a(1 - ab). \quad (5.9)$$

Clamp $a = A \geq 0$ and initialize $b = 0$. Use the downstream *functional coefficient* $q = ab$, whose target is one. Then

$$\dot{q} = \lambda_2 A^2(1 - q), \quad q(t) = 1 - e^{-\lambda_2 A^2 t} \quad (A > 0). \quad (5.10)$$

Therefore

$$\tau_q(\eta | A) = \frac{-\log(1 - \eta)}{\lambda_2 A^2} \quad (A > 0), \quad \tau_q(\eta | 0) = +\infty. \quad (5.11)$$

This is a direct test of hard prerequisite acceleration using a gauge-invariant output coefficient; the intervention remains tied to the specified parameterization. Partial progress in the earlier factor speeds downstream acquisition quadratically in A . Meanwhile, when $b = 0$, the singleton feature still gives $\dot{a} = \lambda_1(1 - a)$, so the seed does not require the later computation.

The effect is not unidirectional at every time: the deeper error also updates a , and $\partial \dot{b} / \partial a = \lambda_2(1 - 2ab)$ changes sign. The hierarchy is in the asymmetric availability of learning signal and the hard intervention, not in a globally positive cross-derivative.

5.4 Full joint training: exact early-time acquisition orders

All layers now train simultaneously, with $a_j(0) = 0$. No layerwise schedule is imposed.

Proposition 5.1 (Staircase onsets). *For unit teacher factors and positive fixed λ_j , the solution has*

$$a_j(t) = C_j t^{\nu_j} + O(t^{\nu_j+1}), \quad \nu_j = 2^{j-1}, \quad (5.12)$$

where

$$C_1 = \lambda_1, \quad C_j = \frac{\lambda_j}{2^{j-1}} \prod_{i < j} C_i = 2^{-(2^{j-1}-1)} \lambda_j \prod_{i < j} \lambda_i^{2^{j-1-i}}. \quad (5.13)$$

Proof. The polynomial vector field has a locally analytic solution. The seed satisfies $\dot{a}_1 = \lambda_1 + O(t)$. If earlier exponents are known, the $m = j$ drive in equation (5.7) has order $\sum_{i < j} \nu_i$. All $m > j$ drives contain at least one additional vanishing factor. The decay term contains squared earlier factors and is higher order. Integrating the leading drive gives $\nu_j = 1 + \sum_{i < j} \nu_i = 2^{j-1}$ and the recurrence for C_j . Induction gives its closed form. $\square$

For example,

$$a_1 \sim \lambda_1 t, \quad a_2 \sim \frac{\lambda_1 \lambda_2}{2} t^2, \quad a_3 \sim \frac{\lambda_1^2 \lambda_2 \lambda_3}{8} t^4. \quad (5.14)$$

The feature-amplitude exponents are 1, 2, 4, 8, ..., whereas the prefix output coefficient p_j has order

$$p_j(t) \sim \left(\prod_{i \leq j} C_i \right) t^{2^j-1}. \quad (5.15)$$

Thus a statement about an onset exponent must specify whether it concerns a constituent amplitude or a composite output.

For $\tau_j(\eta) = \inf\{t : a_j(t) \geq \eta\}$,

$$\tau_j(\eta) = \left(\frac{\eta}{C_j} \right)^{1/2^{j-1}} (1 + o(1)), \quad \eta \downarrow 0. \quad (5.16)$$

Every sufficiently small common threshold is crossed in the order $1, 2, \dots, N$. These are small-threshold asymptotics at fixed finite N , not a theorem about fixed-threshold convergence times, large-depth limits, or a finite interval of exactly zero activity. The source draft establishes the same onset result and records failures of stronger all-threshold ordering claims [P1, Proposition 5 and Section 5.2].

5.5 A quantitative clock-coupling prediction

For $i < j$, changing only the lower-prefix weight from λ_i to $q\lambda_i$, with $q > 0$ fixed, gives

$$\frac{\tau_j(\eta; q\lambda_i)}{\tau_j(\eta; \lambda_i)} \longrightarrow q^{-2^{-i}} \quad (\eta \downarrow 0). \quad (5.17)$$

This is an immediate consequence of equations (5.13) and (5.16). It quantifies how weaker supervision of an earlier feature delays later onsets. A change to another independent channel k' has no effect at all. Unlike mere staging, the within-channel clocks are coupled.

The intervention changes the strength of a term in the teacher–student task; it is complementary to, not identical with, the state-clamping experiment in equation (5.11). Both are useful diagnostics. Neither supports a claim that every late-time learning rate improves monotonically with every lower-prefix weight.

5.6 Beyond a comb and the role of fixed readouts

A branching staircase can read out several subtree functions $H_{v,k} = p_{v,k} \Psi_{S_{v,k}}$, where $p_{v,k} = \prod_{u \in \mathcal{P}(v)} a_{u,k}$ and $\mathcal{P}(v)$ is the set of trained scalar factors below that readout. Distinct node supports are orthogonal, so

$$\mathcal{L} = \frac{1}{2} \sum_{v,k} \lambda_{v,k} (p_{v,k} - p_{v,k}^*)^2, \quad (5.18)$$

with exact flow

$$\dot{a}_{u,k} = \sum_{v: u \in \mathcal{P}(v)} \lambda_{v,k} (p_{v,k}^* - p_{v,k}) \prod_{w \in \mathcal{P}(v) \setminus \{u\}} a_{w,k}. \quad (5.19)$$

This gives a nested partial order when suitable lower subtrees are directly supervised. The comb’s specific exponents should not be transferred to a different set of trainable gates without re-deriving them.

Fixed nonzero readouts are important. Replacing each $\sqrt{\lambda_j}$ by an extra trained factor introduces additional symmetries and can remove the nonzero seed at all-zero initialization. The source staircase theorem is about its stated parameterization, not every possible deep implementation of the same polynomial.

6 Channelwise residual TTNs: soft acceleration with bypasses

6.1 Model and the meaning of “diagonal” with a residual stream

Fix orthogonal leaf features as in proposition 2.2, and initially keep their gains fixed at one. At every internal node, including the root, let

$$h_{v,k} = \rho_v (h_{vL,k} + h_{vR,k}) + \gamma_v b_{v,k} h_{vL,k} h_{vR,k}, \quad h_{i,k} = z_{i,k}. \quad (6.1)$$

Take $f_b = \sum_k h_{o,k}$. The teacher has the same fixed ρ_v, γ_v and leaf features, with $b_{v,k} = \beta_{v,k}$. The trained parameters in this section are precisely the $b_{v,k}$. Unless a limit is being considered, assume $\rho_v, \gamma_v > 0$.

The bilinear term only composes matching channels. The residual term passes an existing feature to its

parent without requiring multiplication by the sibling representation. As in the project proposal, this is a binary identity-like merge, not literally the single-stream update of a transformer [P5, Section 3].

Augment the state by a constant coordinate: $\bar{h} = (1, h_1, \dots, h_r)$. The corresponding ordinary TTN core has nonzero entries

$$\bar{B}_{000} = 1, \quad \bar{B}_{k,k,0} = \bar{B}_{k,0,k} = \rho_v, \quad \bar{B}_{k,k,k} = \gamma_v b_{v,k}. \quad (6.2)$$

Its bond dimension is at most $r + 1$. The entries $(k, k, 0)$ and $(k, 0, k)$ are *not* superdiagonal. Thus “channelwise residual TTN” is the precise name: it preserves separation of nonconstant channels, but is not a strict superdiagonal homogeneous TTN after augmentation. The fixed routing entries must not be trained when interpreting this as the original residual architecture [P2, Section 3.1].

6.2 General parameter gradient flow

Let $e(x) = f_b(x) - f_\star(x)$ and let $p_{v,k}(x) = \partial f_b / \partial h_{v,k}$. With the fixed summation head, $p_{o,k} = 1$. Backpropagation gives

$$p_{vL,k} = p_{v,k}(\rho_v + \gamma_v b_{v,k} h_{vR,k}), \quad (6.3)$$

$$p_{vR,k} = p_{v,k}(\rho_v + \gamma_v b_{v,k} h_{vL,k}), \quad (6.4)$$

and hence

$$\dot{b}_{v,k} = -\gamma_v \mathbb{E}[e p_{v,k} h_{vL,k} h_{vR,k}]. \quad (6.5)$$

If leaf gains were also trained, their equation would be $\dot{a}_{i,k} = -\mathbb{E}[e p_{i,k} z_{i,k}]$; they are held fixed for the next theorems. The displayed equations are the channelwise specialization of the general parameter flow in the residual note [P2, Section 5.3].

6.3 An exact population recursion with linear cost

For a single channel, let $Q_v^{ss} = \mathbb{E}[h_v^2]$, $Q_v^{st} = \mathbb{E}[h_v h_v^\star]$, and $Q_v^{tt} = \mathbb{E}[(h_v^\star)^2]$. All hidden states are centered. Left and right subtrees are independent. Therefore, for $\mu, \nu \in \{s, t\}$,

$$Q_v^{\mu\nu} = \rho_v^2 (Q_{vL}^{\mu\nu} + Q_{vR}^{\mu\nu}) + \gamma_v^2 b_v^\mu b_v^\nu Q_{vL}^{\mu\nu} Q_{vR}^{\mu\nu}, \quad (6.6)$$

where $b_v^s = b_v$ and $b_v^t = \beta_v$. At a fixed unit-feature leaf all three moments equal one. Expanding the product of two copies of equation (6.1) proves the formula: cross terms between left and right linear parts vanish, as do linear–bilinear terms because one child has mean zero.

Under the shared orthogonal leaf-basis assumption, different channels have zero cross-moments. Thus

$$\mathcal{L}_{\text{res}} = \frac{1}{2} \sum_k (Q_{o,k}^{ss} - 2Q_{o,k}^{st} + Q_{o,k}^{tt}). \quad (6.7)$$

Loss evaluation and reverse differentiation of this recursion cost $O(Nr)$ scalar operations, apart from input preprocessing. There is no need to expand all subsets of leaves. This is a computational closure *given the parameters*; it is not a claim that the moments alone satisfy autonomous learning equations. The dense-core analogue in the source uses small matrices rather than scalars [P2, Section 5.2].

6.4 The three-leaf example: exact loss and dynamics

Consider one channel on the tree $((1, 2), 3)$, with common $\rho, \gamma > 0$:

$$u_a = \rho(z_1 + z_2) + \gamma a z_1 z_2, \quad f_{a,b} = \rho(u_a + z_3) + \gamma b u_a z_3. \quad (6.8)$$

The teacher uses $a = \alpha > 0, b = \beta > 0$. Expanding,

$$f_{a,b} = \rho^2(z_1 + z_2) + \rho z_3 + \rho \gamma a z_1 z_2 \quad (6.9)$$

$$+ \rho \gamma b(z_1 z_3 + z_2 z_3) + \gamma^2 a b z_1 z_2 z_3. \quad (6.10)$$

The fixed linear terms cancel in the error. Distinct supports are orthogonal, so

$$\mathcal{L}(a, b) = \frac{u}{2}(a - \alpha)^2 + \frac{v}{2}(b - \beta)^2 + \frac{w}{2}(ab - \alpha\beta)^2, \quad (6.11)$$

where

$$u = \rho^2 \gamma^2, \quad v = 2\rho^2 \gamma^2, \quad w = \gamma^4. \quad (6.12)$$

There are two orthogonal upper pairwise terms, which explains the factor two in v . Differentiation gives

$$\begin{aligned} \dot{a} &= u(\alpha - a) + wb(\alpha\beta - ab), \\ \dot{b} &= v(\beta - b) + wa(\alpha\beta - ab). \end{aligned} \quad (6.13)$$

Both core parameters can learn directly, and they share the triple interaction.

Source and status. Equations (6.8) and (6.13) and the following two-parameter results reproduce and explain Section 6 of the residual-TTN note [P2]. They are not additional independent evidence beyond that model.

6.5 A proved soft acceleration effect

The observable b is identifiable from accessible pairwise coefficients:

$$a = \frac{\mathbb{E}[f_{a,b} z_1 z_2]}{\rho \gamma}, \quad b = \frac{\mathbb{E}[f_{a,b}(z_1 z_3 + z_2 z_3)]}{2\rho \gamma}. \quad (6.14)$$

Thus b is not merely an arbitrary hidden coordinate. It measures a particular upper composition and its output features.

Proposition 6.1 (Soft upper-core acceleration). *Fix $b(0) = b_0 \in [0, \eta\beta]$ and compare two interventions: clamp $a = 0$, or clamp $a = \alpha$. Then*

$$b_{\text{blocked}}(t) = \beta - (\beta - b_0)e^{-vt}, \quad (6.15)$$

$$b_{\text{learned}}(t) = \beta - (\beta - b_0)e^{-(v+w\alpha^2)t}, \quad (6.16)$$

so

$$\frac{\tau_b(\eta \mid a = 0)}{\tau_b(\eta \mid a = \alpha)} = 1 + \frac{w\alpha^2}{v} = 1 + \frac{\gamma^2 \alpha^2}{2\rho^2} > 1. \quad (6.17)$$

Both acquisition times are finite.

Proof. At $a = 0$, equation (6.13) becomes $\dot{b} = v(\beta - b)$. At $a = \alpha$, it becomes $\dot{b} = (v + w\alpha^2)(\beta - b)$.

Solve the two linear equations and impose $b = \eta\beta$. The common logarithmic factor is $\log[(\beta - b_0)/((1 - \eta)\beta)]$. $\square$

This is acceleration of an upper composition by a learned lower composition, with a bypass. But it is not a theorem of universal directed hierarchy. In joint training,

$$\dot{a}|_{(0,0)} = u\alpha, \quad \dot{b}|_{(0,0)} = v\beta, \quad \frac{\partial \dot{b}}{\partial a} = w(\alpha\beta - 2ab). \quad (6.18)$$

Both parameters start immediately. Either can have the earlier acquisition time, and the local influence can reverse sign. The reciprocal clamping experiment can also accelerate a .

6.6 What is being reused: the changing tangent feature

The function seen by the upper parameter is

$$\chi_b(x; a) := \frac{\partial f_{a,b}}{\partial b} = \rho\gamma(z_1 z_3 + z_2 z_3) + \gamma^2 a z_1 z_2 z_3. \quad (6.19)$$

It changes during training: a lower composition adds a new degree-three component to the upper parameter's tangent feature. Its squared norm is

$$\|\chi_b\|_{L^2}^2 = v + wa^2, \quad (6.20)$$

and the parameter tangent-feature Gram matrix is

$$\mathbb{E} [\nabla_{(a,b)} f \nabla_{(a,b)} f^\top] = \begin{pmatrix} u + wb^2 & wab \\ wab & v + wa^2 \end{pmatrix}. \quad (6.21)$$

The learned representation changes the upper optimization problem; no time-dependent learning-rate schedule was inserted. An increased tangent norm alone would not prove acceleration, because the residual forcing also matters. The clamped teacher-aligned calculation proves it in this case.

The triple output coefficient is $\gamma^2 ab$. It remains zero when a is blocked, regardless of how well b learns. Thus *soft learnability of the upper core does not imply that the full triple capability has a bypass*. A bypass for that complete function would require another route capable of producing the same three-leaf interaction.

6.7 Joint convergence in this particular example

Proposition 6.2 (Two-core convergence). *For $u, v, w, \alpha, \beta > 0$, the joint flow equation (6.13) starting in the nonnegative quadrant is global and converges to (α, β) .*

Proof. At $a = 0$ or $b = 0$, the corresponding derivative points strictly inward, so the quadrant is invariant. The positive separate quadratic terms in equation (6.11) make loss sublevel sets compact, and $\dot{\mathcal{L}} = -\dot{a}^2 - \dot{b}^2$. Hence the trajectory is bounded and global. At an interior stationary point set $C = ab - \alpha\beta$. If $C > 0$, stationarity forces both $a < \alpha$ and $b < \beta$, a contradiction. If $C < 0$, it forces both inequalities in the opposite direction, again a contradiction. Thus $C = 0$ and $(a, b) = (\alpha, \beta)$. The compact limit set of the gradient flow contains only this stationary point. $\square$

This convergence result does not impose an ordering of acquisition, and the proof should not be transplanted to arbitrary residual trees without checking their stationary points.

6.8 A support expansion for arbitrary residual trees

The three-leaf mechanism generalizes cleanly with fixed leaves. Work in one channel. For a nonempty leaf set S , define

$$\mathcal{A}(S) = \{v : S \cap S_{vL} \neq \emptyset, S \cap S_{vR} \neq \emptyset\}, \quad (6.22)$$

$$\mathcal{U}(S) = \{v : \text{exactly one of } S \cap S_{vL}, S \cap S_{vR} \text{ is nonempty}\}, \quad (6.23)$$

$$d_S = \prod_{v \in \mathcal{U}(S)} \rho_v \prod_{v \in \mathcal{A}(S)} \gamma_v. \quad (6.24)$$

Here $\mathcal{A}(S)$ is the set of nodes where the selected support branches into both children. At a node with only one selected child, the residual term propagates that support.

Proposition 6.3 (Residual support coefficients). *The coefficient of $\Psi_S = \prod_{i \in S} z_i$ in the root output is*

$$c_S(b) = d_S \prod_{v \in \mathcal{A}(S)} b_v. \quad (6.25)$$

Consequently,

$$\mathcal{L}_{\text{res}}(b) = \frac{1}{2} \sum_{\emptyset \neq S \subseteq [N]} d_S^2 \left(\prod_{v \in \mathcal{A}(S)} b_v - \prod_{v \in \mathcal{A}(S)} \beta_v \right)^2. \quad (6.26)$$

Singleton supports have empty products and contribute zero error.

Proof. At a leaf the statement is immediate. At an internal node, a support contained in just one child can arise only from that child's residual summand, gaining ρ_v . A support intersecting both children can arise only from the bilinear term, gaining $\gamma_v b_v$ times the two child coefficients. Induction gives equation (6.25); orthogonality gives equation (6.26). $\square$

For a pair $S = \{i, j\}$, $\mathcal{A}(S)$ contains only the lowest common ancestor $v = \text{LCA}(i, j)$. Therefore every core has a directly supervised pairwise coefficient. Define

$$\mu_v = \sum_{\substack{i \in S_{vL} \\ j \in S_{vR}}} d_{\{i,j\}}^2 > 0. \quad (6.27)$$

Then the loss contains $\mu_v(b_v - \beta_v)^2/2$ for every internal v , in addition to higher-order shared-factor terms. In particular,

$$\dot{b}_v(0) = \mu_v \beta_v \quad (6.28)$$

when all cores start at zero. When every teacher core is nonzero, all core amplitudes have an order- t onset in this matched residual teacher family. A strict staircase of Taylor orders is absent, despite possible large differences in rates.

6.9 An arbitrary-tree soft acceleration theorem

Theorem 6.4 (Conditional soft facilitation in a residual tree). *Assume positive fixed ρ_v, γ_v and positive teacher cores β_v . Choose a free core b_v and a set $\mathcal{B}$ of other cores to block. Hold $b_u = 0$ for $u \in \mathcal{B}$ and $b_u = \beta_u$ for every other $u \neq v$. Then*

$$\dot{b}_v = \kappa_v(\mathcal{B})(\beta_v - b_v), \quad (6.29)$$

where

$$\kappa_v(\mathcal{B}) = \sum_{\substack{S: v \in \mathcal{A}(S) \\ \mathcal{A}(S) \cap \mathcal{B} = \emptyset}} d_S^2 \prod_{u \in \mathcal{A}(S) \setminus \{v\}} \beta_u^2 \geq \mu_v > 0. \quad (6.30)$$

Releasing the blocked cores to their teacher values increases the rate to $\kappa_v(\emptyset)$. The increase is strict whenever at least one supported interaction contains both v and a blocked core. For a fixed $b_v(0) \in [0, \eta\beta_v)$,

$$\frac{\tau_v(\eta \mid \mathcal{B} \text{ blocked})}{\tau_v(\eta \mid \mathcal{B} \text{ learned})} = \frac{\kappa_v(\emptyset)}{\kappa_v(\mathcal{B})}. \quad (6.31)$$

Both times are finite.

Proof. In equation (6.26), a term whose active set intersects $\mathcal{B}$ has student coefficient zero and zero derivative with respect to b_v . A term containing v but no blocked node reduces, up to its positive prefactor, to $(b_v - \beta_v)^2/2$. Summing these prefactors gives equation (6.30). The pairwise terms with active set $\{v\}$ always survive, proving the lower bound. Restoring blocked cores adds nonnegative terms, with a positive addition under the stated condition. Solving the scalar linear flow gives the time ratio. $\square$

Interpretation and scope. This is a derived extension of the source’s three-leaf calculation. If the blocked core is below v , the result is a concrete hierarchical reuse intervention. But the theorem deliberately freezes all other cores; it does not assert the same ordering or comparison along unconstrained joint training. Other interacting cores can facilitate each other too. Moreover, the full teacher cannot generally be represented with the blocked cores held at zero. The theorem concerns the identifiable core b_v and its surviving output features, not zero full-task loss under the intervention.

6.10 Advantages and limits of soft hierarchy

Soft acceleration avoids an absolute bottleneck: an upper operation can make progress from a residual representation even before a particular lower composition is acquired. It also permits overlapping learning phases rather than forcing a strict curriculum. In the conditional theorem, the surviving pairwise route provides a finite baseline rate μ_v , and learned composites add to it. These are concrete advantages of the model, not a theorem that residual architectures always optimize faster at fixed model size or task difficulty.

There is a corresponding modeling bias. Positive residual routes and fixed correct leaf directions supply every core with low-order teacher signal. Hard targets with absent low-order correlations are not represented by this favorable matched family in the same way. The theory should therefore be tested after weakening those signals, learning the leaf directions, or allowing competing paths.

The limit $\rho \rightarrow 0$ in the three-leaf example gives $u = v = 0$ and leaves only $(ab - \alpha\beta)^2$. It recovers *mutual product gating*, not the asymmetric staircase model. A staircase retains a lower-order seed while removing a downstream direct route; vanishing all residual paths removes both.

7 A unified comparison of the three mechanisms

7.1 The smallest common objective

The two-factor family

$$\mathcal{L}_{A,B,AB}(a, b) = \frac{\lambda_A}{2}(a - \alpha)^2 + \frac{\lambda_B}{2}(b - \beta)^2 + \frac{\lambda_{AB}}{2}(ab - \alpha\beta)^2 \quad (7.1)$$

summarizes the distinction. It is an objective-level comparison; a given architecture can constrain which weights are realizable.

Regime	Weights	Learning mechanism
Pure product	$\lambda_A = \lambda_B = 0, \lambda_{AB} > 0$	Symmetric factor gating. No independently anchored lower feature.
Staircase	$\lambda_A > 0, \lambda_B = 0, \lambda_{AB} > 0$	The lower feature has a seed; the upper factor needs it. Hard prerequisite structure.
Residual-like	$\lambda_A, \lambda_B, \lambda_{AB} > 0$	Each factor has a direct route and both share a composite. Soft, potentially reciprocal facilitation.

Clamping $a = 0$ versus $a = \alpha$ gives downstream relaxation rates λ_B and $\lambda_B + \lambda_{AB}\alpha^2$. Thus λ_B is a mathematically explicit leak or bypass parameter. The staircase is the singular case where the blocked rate vanishes while the lower seed remains.

7.2 Shared coefficients, not dense channel mixing, are enough

Let $c(\theta)$ collect orthonormal output coefficients and $J = \partial c / \partial \theta$. Then

$$\dot{c} = JJ^\top (c^\star - c). \quad (7.2)$$

A diagonal architecture can therefore have coupled coefficient dynamics even when the function-space loss is an orthogonal sum. The source of coupling is a shared parameter in different rows of J .

In a homogeneous channel, there is only one output coefficient. In the staircase there are nested prefix coefficients. In the residual tree there is a coefficient for every nonempty support, with sharing prescribed by the pruning sets $\mathcal{A}(S)$. These mechanisms are captured by monomial parameterizations

$$c_\alpha(\theta) = d_\alpha \prod_p \theta_p^{M_{\alpha p}}, \quad (7.3)$$

where M records which parameters a feature coefficient uses. Here M has only zero–one entries; other polynomial models can involve repeated powers. This gives a useful connection to the feature ODE discussed in Section 10.3.

7.3 A fair comparison must control expressivity

A pure N -leaf homogeneous TTN cannot fit the lower-degree terms of a generic residual teacher. A comb staircase also omits many non-prefix interactions. Therefore training all three against the same unrestricted residual teacher and comparing final loss mixes optimization with misspecification. For mechanistic tests, use matched realizable subproblems, compare selected common coefficients, or embed the architectures in a common model with controlled readout and bypass gates. Report the irreducible approximation error separately.

Likewise, duplicating independent diagonal channels does not create cross-channel hierarchical composition. It creates several copies of the same within-channel mechanism. Block-diagonal or sparse non-diagonal cores are a natural next model when the question is how distinct lower-level representations get combined.

8 From amplitudes to genuine direction learning

8.1 An exact directional staircase extension

The primitive directions need not remain fixed to retain all tractability. For one channel, replace $a_j z_j$ by $w_j^\top x_j$, with unit-norm teacher direction u_j . Define

$$f_W(x) = \sum_m \sqrt{\lambda_m} \prod_{j \leq m} (w_j^\top x_j), \quad f_\star(x) = \sum_m \sqrt{\lambda_m} \prod_{j \leq m} (u_j^\top x_j). \quad (8.1)$$

Set $a_j = w_j^\top u_j$ and $q_j = \|w_j\|^2$. Independence and whitening give

$$\mathcal{L}(W) = \frac{1}{2} \sum_m \lambda_m \left(\prod_{j \leq m} q_j - 2 \prod_{j \leq m} a_j + 1 \right). \quad (8.2)$$

With

$$D_j = \sum_{m \geq j} \lambda_m \prod_{i \leq m, i \neq j} q_i, \quad B_j = \sum_{m \geq j} \lambda_m \prod_{i \leq m, i \neq j} a_i, \quad (8.3)$$

the full vector gradient flow and its exact overlap closure are

$$\dot{w}_j = -D_j w_j + B_j u_j, \quad \dot{a}_j = -D_j a_j + B_j, \quad \dot{q}_j = -2D_j q_j + 2B_j a_j. \quad (8.4)$$

The perpendicular energy satisfies

$$\frac{d}{dt}(q_j - a_j^2) = -2D_j(q_j - a_j^2). \quad (8.5)$$

Thus $2N$ scalar variables describe the learning of unknown leaf directions as well as their norms, independently of the ambient block dimensions once their initial overlaps are specified. The teacher-directed term still contains the product of prerequisite alignments. This is the cofactor structure of the project's rank-one staircase model [P1, Section 4.3 and Proposition 4].

This extension should not be misdescribed as the fully fixed-basis diagonal model. Its cores still perform channelwise multiplication, but its leaf directions move. For multiple freely moving channels, their mutual overlaps generally create additional couplings. Independent copies can be preserved by constraining channels to preassigned orthogonal input subspaces; that restriction learns directions *within* each subspace, not the correct grouping of directions across channels.

8.2 Directional soft facilitation in a residual motif

For independent whitened vector blocks x, y , consider

$$F_{w,z} = \rho(w^\top x + z^\top y) + \gamma(w^\top x)(z^\top y), \quad (8.6)$$

with teacher vectors $w^\star, z^\star$. Its exact loss is

$$\mathcal{L} = \frac{\rho^2}{2} (\|w - w^\star\|^2 + \|z - z^\star\|^2) + \frac{\gamma^2}{2} \|w \otimes z - w^\star \otimes z^\star\|_F^2, \quad (8.7)$$

and

$$\dot{z} = [\rho^2 + \gamma^2(w^\top w^\star)]z^\star - [\rho^2 + \gamma^2\|w\|^2]z. \quad (8.8)$$

Clamping $w = 0$ or $w = w^\star$ gives rates ρ^2 and $\rho^2 + \gamma^2\|w^\star\|^2$ toward the same vector $z^\star$.

A stricter directional control compares $w = w^\star$ with $w = w_\perp$ of the same norm but orthogonal to $w^\star$. Put $\kappa = \rho^2 + \gamma^2 \|w^\star\|^2$. For the same $z(0)$,

$$z_{\text{aligned}}(t) - z_{\text{orthogonal}}(t) = \frac{\gamma^2 \|w^\star\|^2}{\kappa} (1 - e^{-\kappa t}) z^\star. \quad (8.9)$$

Both runs have the same decay of perpendicular components; the learned direction supplies more target-aligned forcing. With a nonnegative initial target projection and a nonzero perpendicular component, this improves directional alignment. This source example establishes mutual directional facilitation, not a multi-level directed hierarchy [P2, Section 6.4].

8.3 The remaining representational restriction

Even with learned leaf directions, a strictly channel-diagonal homogeneous core sends (k, k) to k . It cannot discover that left feature i and right feature j should combine into an independently represented parent feature k . A block-diagonal TTN allows such mixing inside small groups while keeping different groups independent. Sparse routing offers a similar compromise. These are proposed extensions, not consequences of the scalar proofs.

An aligned or orthogonal family imposed as a constraint is also different from a family preserved by unrestricted gradient flow. Claims about the latter require a proof for the exact architecture, readout, data metric, and optimizer. A time-dependent change of basis or arbitrary bond gauge transformation does not by itself preserve Euclidean gradient dynamics [P2, Section 7].

9 How to extract and track the learned features

9.1 Known basis: recover coefficients directly

When the $\Psi_{S,k}$ are known, the most transparent observable is

$$c_{S,k}(t) = \mathbb{E}[f_{\theta(t)}(x) \Psi_{S,k}(x)]. \quad (9.1)$$

It can be computed analytically or estimated on a held-out sample. A Monte Carlo estimator is unbiased when the expectation exists; a standard root-sample-size error statement additionally needs finite variance of the product being averaged.

For the pure diagonal model, $c_{[N],k} = s_k$. For the staircase,

$$p_{j,k} = \frac{\mathbb{E}[f_a \Psi_{[j],k}]}{\sqrt{\lambda_{j,k}}}, \quad a_{1,k} = p_{1,k}, \quad a_{j,k} = \frac{p_{j,k}}{p_{j-1,k}} \quad (p_{j-1,k} \neq 0). \quad (9.2)$$

The prefix coefficients fix the multiplicative ambiguity wherever these ratios are defined. Near zero, ratios are ill-conditioned; one should report the coefficients and their uncertainty rather than declare a noisy ratio to be a discovered feature.

For the fixed-leaf residual model, collect the pairwise feature associated with node v :

$$G_{v,k}(x) = \sum_{\substack{i \in S_{vL} \\ j \in S_{vR}}} d_{\{i,j\}} z_{i,k} z_{j,k}, \quad \mathbb{E}[G_{v,k}^2] = \mu_v. \quad (9.3)$$

All higher-support terms are orthogonal to $G_{v,k}$, and different nodes correspond to disjoint sets of leaf

pairs. Hence

$$b_{v,k} = \frac{\mathbb{E}[f_b G_{v,k}]}{\mu_v}. \quad (9.4)$$

Equivalently, $G_{v,k}$ is the tangent feature $\partial f / \partial b_{v,k}$ at the all-zero core state. This gives an explicit, function-based way to extract the core strength without interpreting an arbitrary hidden coordinate. The formula depends on the fixed known leaf features and residual routing; it is not an architecture-independent method for arbitrary neural networks.

9.2 Unknown basis: function-space SVD across tree cuts

For independent input groups x_S, x_{S^c} , any finite-rank represented function has a Schmidt decomposition

$$f_\theta(x) = \sum_j \sigma_{S,j}(\theta) \phi_{S,j}(x_S) \psi_{S,j}(x_{S^c}), \quad (9.5)$$

where the left and right functions are orthonormal in their respective population L^2 spaces. The singular values and subspaces are invariant to internal TTN gauge changes that leave f_θ unchanged. This is the feature definition proposed in the teacher–student program [P5, Section 4] and refined in the residual note [P2, Section 7].

With feature columns Φ_S, Φ_{S^c} and coefficient matrix A , write $f = \Phi_S^\top A \Phi_{S^c}$. If $G_S = \mathbb{E}[\Phi_S \Phi_S^\top]$ and $G_{S^c} = \mathbb{E}[\Phi_{S^c} \Phi_{S^c}^\top]$, the matrix to decompose is

$$M_S = G_S^{1/2} A G_{S^c}^{1/2}. \quad (9.6)$$

Both side metrics are needed. Zero-variance directions should be removed before whitening. A bare local-core SVD is not the same calculation because it omits the subtrees and the complementary environment.

For a TTN, orthonormalize the subtree and complement bases toward the chosen edge, form the resulting small center matrix, and compute its SVD. This avoids forming the full physical tensor. A truncation at one cut has squared L^2 error $\sum_{j>q} \sigma_{S,j}^2$; simultaneous hierarchical truncation has more qualified, generally quasi-optimal guarantees [2]. The augmented residual representation has cut rank at most $r + 1$, but its singular values are not simply the individual $b_{v,k}$.

9.3 A worked residual cut: one hidden channel can change its function

For equation (6.8), cut $\{1, 2\} \mid \{3\}$. Let

$$Q(a) = \mathbb{E}[u_a^2] = 2\rho^2 + \gamma^2 a^2. \quad (9.7)$$

In the orthonormal left basis $\{1, u_a / \sqrt{Q(a)}\}$ and right basis $\{1, z_3\}$, the exact cut matrix is

$$M(a, b) = \begin{pmatrix} 0 & \rho \\ \rho \sqrt{Q(a)} & \gamma b \sqrt{Q(a)} \end{pmatrix}. \quad (9.8)$$

Its singular values are readily computed from a 2×2 SVD. The normalized function $u_a / \sqrt{Q(a)}$ changes from a linear combination of z_1, z_2 toward a mixture containing $z_1 z_2$. Thus scalar core learning can change the direction of a *composite function* within a known low-dimensional function space. This is richer than a homogeneous channel’s fixed normalized product, but still not discovery of an unknown high-dimensional

primitive direction.

The canonical singular functions generally mix the constant function and u_a ; they need not coincide with a named hidden neuron. They also depend on the environment through b . This example makes concrete why local core weights, hidden features, and cut singular features should be tracked separately.

9.4 Alignment, amplification, and effective rank

For a chosen, well-defined rank- q singular subspace, let $P_S(t)$ and $P_S^\star$ be its student and teacher orthogonal projectors. A subspace-alignment observable is

$$A_S^{(q)}(t) = \frac{1}{q} \text{tr}(P_S(t)P_S^\star). \quad (9.9)$$

Use subspaces rather than individual vectors at repeated singular values, and report rank deficiencies instead of filling missing directions arbitrarily. Track the singular values themselves to distinguish alignment from amplification. An effective rank, for example

$$r_{\text{eff}}(S, t) = \frac{\left(\sum_j \sigma_{S,j}^2\right)^2}{\sum_j \sigma_{S,j}^4}, \quad (9.10)$$

is meaningful when the denominator is nonzero, but it is a summary of a spectrum, not evidence of a causal hierarchy.

For a fixed-basis pure diagonal TTN, any already active matching teacher mode is aligned from the outset; the nontrivial dynamics is in its strength. Observing its amplitude cross a threshold should not be reported as the first discovery of its direction. For a residual network, alignment of a subtree function can evolve, as [equation \(9.8\)](#) illustrates.

9.5 Tangent features and task-directed signal

A useful complement to cut features is the family

$$\chi_p(x; t) = \frac{\partial f_{\theta(t)}(x)}{\partial \theta_p}, \quad K_{pq}(t) = \mathbb{E}[\chi_p \chi_q], \quad s_p(t) = \mathbb{E}[(f_\star - f_\theta)\chi_p]. \quad (9.11)$$

Here $s_p = \dot{\theta}_p$ under the chosen gradient flow. The Gram matrix measures the strength and overlap of directions available to training; s_p measures which of them the current task drives. Both are needed. For the residual example, [equation \(6.19\)](#) explicitly shows how the lower computation adds a previously absent nonlinear component to the upper tangent feature.

Output-function SVD can miss a hidden feature whose readout is currently zero. Representation-level probes can detect such a dormant feature, but they depend more strongly on the architecture. Reporting output features, hidden representations, and tangent features together separates what is represented, what influences the prediction, and what is currently learnable.

9.6 A practical causal test

At several checkpoints, identify a lower feature using a fixed teacher reference or a clearly specified subspace. Compare a blocked and a teacher-aligned intervention with the same downstream initialization and learning rates. Preserve the lower feature's norm when the test is meant to isolate direction. Ensure that the intervention does not itself raise the downstream observable above the threshold before any training

occurs. The staircase comparison used $b(0) = 0$ to avoid this problem; the residual comparison uses a pairwise coefficient of b that is unaffected instantaneously by changing a .

Measure acquisition times at several thresholds, not only one. Repeat the intervention in the reverse direction to distinguish reciprocal facilitation from a directed prerequisite. Monitor the whole output error as well as the selected feature, since a blocked run may learn an upper parameter yet remain unable to express a higher-order capability. Joint training curves provide context, but the intervention is what tests whether the earlier learned computation changes the subsequent learning process.

10 Relation to the literature review and a research program

10.1 What the review motivates, and what the diagonal model supplies

The literature review separates expressive power, learnability, and training dynamics, and argues for a theory with interpretable, gauge-aware order parameters under joint training [P3]. The present models address a specific part of that program: exact equations for amplification, reuse, and acquisition clocks in a planted structured problem. They do not settle generic hierarchical direction discovery or the emergence of human-interpretable concepts.

The review’s suggestion to move from bond dimension one to diagonal or orthogonally structured cores is productive, but requires a refinement. Strictly compatible superdiagonal homogeneous cores with fixed leaf bases still give the CP collapse in [equation \(2.11\)](#). Increasing width alone does not restore independent hierarchical spectra. To obtain richer dynamics while retaining tractability, one must add shared lower-order readouts, residual routing, moving directions, or controlled mixing between channels. Orthogonally decomposable local cores whose bases are *not* compatibly aligned are a different, less restrictive model and need not have this collapse.

10.2 The relevant theoretical connections

Deep linear learning dynamics. Saxe, McClelland, and Ganguli analyze nonlinear optimization dynamics despite a linear input–output map, with exact aligned-mode solutions under controlled assumptions [1]. The pure diagonal TTN inherits the same amplitude equations in the regime specified here. Its value is therefore as a calibration baseline. The staircase and residual extensions ask what changes when a learned factor influences several distinguishable nonlinear coefficients, rather than merely one singular-mode strength.

Expressivity through tensor factorization. Cohen, Sharir, and Shashua connect deep arithmetic circuits to hierarchical tensor representations and establish depth-efficiency results for their model classes [3]. Those results concern representational power, not a guarantee that gradient descent discovers the factorization. The strict diagonal homogeneous subclass studied here already has the CP form [equation \(2.11\)](#); it should not be credited with the full expressive advantage of a generic hierarchical factorization.

Staircase and leap complexity. The staircase, merged-staircase, and leap-complexity papers connect target structure to the ability of gradient-based learning to acquire higher-complexity features from simpler ones in their respective Boolean or Gaussian settings [4, 5, 6]. Our prefix model makes a related gating mechanism explicit through shared factors. Its Taylor orders and population hitting times are *not* those papers’ high-dimensional sample-complexity bounds: the input geometry, architecture, and asymptotic questions differ.

Successive dimensionality reduction. Dandi, Pesce, Zdeborová, and Krzakala study hierarchical Gaussian targets and show how learned representations can successively reduce the effective search dimension [7]. Their rigorous results use controlled, layerwise training, while joint backpropagation is investigated numerically. A fixed-direction diagonal toy model does not reproduce their unknown-subspace recovery problem. Its contribution is instead to expose joint-time coupling and intervention responses with explicit formulas; the directional and block-structured extensions are needed to approach the richer mechanism.

Neural low-degree filtering. Neural LoFi formalizes a stylized spectral procedure in which each layer selects accessible low-degree label correlations in the current representation [8]. The residual tangent feature in equation (6.19) gives a concrete small-scale example of a learned representation changing which nonlinear correlation an upper parameter can use. This is a conceptual connection, not an equivalence to the LoFi algorithm or a proof that ordinary end-to-end training follows its layerwise limit.

Hierarchical low-rank bias and difficult targets. Razin, Maman, and Cohen analyze implicit regularization toward low hierarchical tensor rank in hierarchical factorization [9]. Furman, Wäldchen, Bei, and Hodgkinson show that general TTNs can contain hard-to-learn targets while having a conditional benign-landscape property: every minimum-norm local minimum is global, with high-order rank-deficient saddles providing a mechanism for difficulty [10]. Neither result implies that the fixed-basis diagonal models here discover arbitrary compositions. Their relevance is to motivate rank and saddle observables once the model is enlarged beyond globally aligned product channels.

Mean-field and dynamical mean-field theories. Bordelon and Pehlevan give a self-consistent dynamical description of evolving kernels in wide feature-learning networks [11]. This is a different level of theory from the finite scalar closure here. Independent diagonal channels supply a solvable reference distribution of clocks; a meaningful next question is which additional overlaps and response functions are needed when weak channel mixing or moving features are introduced. No general TTN mean-field closure is proved by the moment recursion equation (6.6).

10.3 The direct connection to the attached feature ODE

The attached paper by Turnbull, Ottlik, and Simon proposes a feature ODE for deep polynomial regression, with an effective loss of the form

$$\mathcal{L}_{\text{feature}} = \frac{1}{2} \sum_{\alpha} \left(v_{\alpha} - \tilde{c}_{\alpha} b_{\alpha} \prod_i a_i^{\alpha_i} \right)^2. \quad (10.1)$$

Its shared primitive factors a_i affect several output coefficients. That is the same algebraic organizing principle as equation (7.3): progress in one coefficient can train a factor subsequently reused elsewhere. Staircase TTNs choose nested support sets; residual TTNs choose tree-pruning support sets; the feature ODE organizes sharing by polynomial multi-indices [P4, Section 2 and Appendix C].

The evidential status differs. Our independent-block losses and their stated gradient flows are exact for the restricted models. The attached paper motivates reductions using assumptions on Gaussian data and target structure, and reports approximate agreement with shallow MLP training; it explicitly does not derive a general exact MLP theory. In particular, its empirical success is motivation to compare these TTN ODEs with less constrained networks, not permission to assert that they already predict those networks.

10.4 What these models are useful for

They provide a controlled separation of three explanations for apparent feature ordering: independent mode strengths, symmetric multiplicative gating, and shared-coefficient hierarchical facilitation. They also produce explicit observables rather than requiring interpretation of a loss curve alone: prefix coefficients, residual pairwise core strengths, cut singular subspaces, and time-dependent tangent features. The staircase model gives joint-training onset exponents and coupled clock ratios. The residual model gives finite blocked learning rates and exact conditional acceleration factors.

At the same time, they are deliberately favorable planted problems. The tree is given, the scalar analysis is aligned with the teacher, and the residual family provides direct low-order signals. They do not yet explain how a network identifies which primitives should be composed, whether an alternative circuit is learned, or how acquisition thresholds scale with ambient dimension and finite data.

10.5 A concrete progression beyond the solvable baseline

First: validate mechanisms, not just fitted loss curves. Implement the exact ODEs and the corresponding networks on independent blocks. Match trainable factors, initialization, and learning rates. Test the staircase clock ratio [equation \(5.17\)](#), the residual time ratio [equation \(6.17\)](#), and the core extraction formula [equation \(9.4\)](#). Then compare with empirical gradient descent while measuring the finite-sample departure from orthogonal coefficient loss.

Second: learn directions while keeping the grouping simple. Use the exact rank-one overlap equations [equation \(8.4\)](#) and the equal-norm directional intervention [equation \(8.9\)](#). Extend to multiple constrained orthogonal groups before allowing arbitrary inter-channel overlap. This isolates orientation learning from changes in scalar amplitude or parameter scale.

Third: add controlled composition choices. Replace superdiagonal cores by small block-diagonal cores, sparse learned routes, or a diagonal core plus a measured perturbation. Track cross-channel tangent terms and teacher–student subspace overlaps. Check whether the diagonal family is invariant under the intended unconstrained flow; otherwise quantify departure rather than treating alignment as an automatic property.

Fourth: test robustness and transfer. Introduce anisotropic covariance, correlated leaves, finite-sample noise, alternative optimizers, and architecture mismatch one at a time. A non-whitened metric or correlated feature Gram matrix can itself create coupling; this must be separated from representation reuse. In particular, setting all leaves equal to the same input is not a benign substitution: different supports can collapse to the same polynomial and moment factorization fails. The source’s exact lifting of a sequential residual network uses this *input diagonal*, which is unrelated to superdiagonal cores [P2, Sections 3–4 and 9].

A later link to scaling laws could aggregate many measured feature-acquisition clocks, but the fixed-basis population models alone do not predict realistic pretraining scaling laws. The critical additions are unknown-feature discovery, statistical detectability, and transfer of the intervention predictions to wider or less structured students. This is the research direction proposed by the literature review, not a result established here [P3].

11 Conclusions

A homogeneous diagonal TTN is nonlinear as a function of raw inputs, but its fixed-orthogonal-basis amplitude training is exactly a collection of scalar DLN product problems. It can exhibit mutual factor acceleration; it does not by itself furnish a distinct directed hierarchy of learned nonlinear features. Its

identical cut spectra make this limitation visible geometrically.

A staircase diagonal TTN adds nested observable coefficients with shared factors and fixed nonzero lower readouts. This yields hard prerequisite gating and provably coupled small-threshold onsets under full simultaneous gradient flow. A channelwise residual TTN gives each upper operation an independent lower-order route while allowing learned lower compositions to contribute additional signal. Its soft acceleration is provable for identified observables, including a conditional arbitrary-tree theorem, but a directed training order and bypass of the full composite target do not follow automatically.

The resulting theoretical program is useful precisely because it separates these mechanisms. Fixed-channel models establish how sharing changes acquisition clocks. Directional and block-structured extensions address which representations are learned. Function-space coefficients, cut SVDs, and tangent features make those claims measurable, while carefully designed interventions determine whether the observed sequence is causally hierarchical rather than merely staged.

A Numerical illustrations and verification scope

The source bundle includes `verify_models.py`, `make_figures.py`, and a machine-readable verification report. The checks differentiate the product and staircase losses symbolically; verify product balance; compute the four-stage staircase Taylor coefficients through the required orders; verify the three-leaf residual expansion and gradient; and check the support and conditional-relaxation formulas on a four-leaf binary tree. They supplement, rather than replace, the proofs. They do not validate literature claims, generalization to untested architectures, or the existence of an unconstrained invariant diagonal manifold.

A.1 Simultaneous staircase training

Figure 1 integrates equation (5.5) for three stages, unit weights and teacher factors, and $a(0) = 0$. All coordinates train at once. The solver uses relative tolerance 2×10^{-11} and absolute tolerance 10^{-13} . The verification script also checks the initial terms $a_1 = t + O(t^2)$, $a_2 = t^2/2 + O(t^3)$, $a_3 = t^4/8 + O(t^5)$, and $a_4 = t^8/128 + O(t^9)$.

A.2 Residual intervention trajectories

Take $\rho = 0.5$, $\gamma = 1$, $\alpha = 1.5$, $\beta = 1$, and $b_0 = 0$. The blocked and learned rates are respectively 0.5 and 2.75, so the acquisition-time ratio is 5.5 at every fixed fractional threshold. At $\eta = 0.8$,

$$\tau_{\text{blocked}} = 3.2188758 \dots, \quad \tau_{\text{learned}} = 0.5852501 \dots \quad (\text{A.1})$$

Figure 2 plots the exact exponentials, not a fit to joint training. A separate numerical integration of the joint two-core flow from zero approaches (1.5, 1), consistent with the proved convergence result; it is not used as a substitute for the intervention comparison.

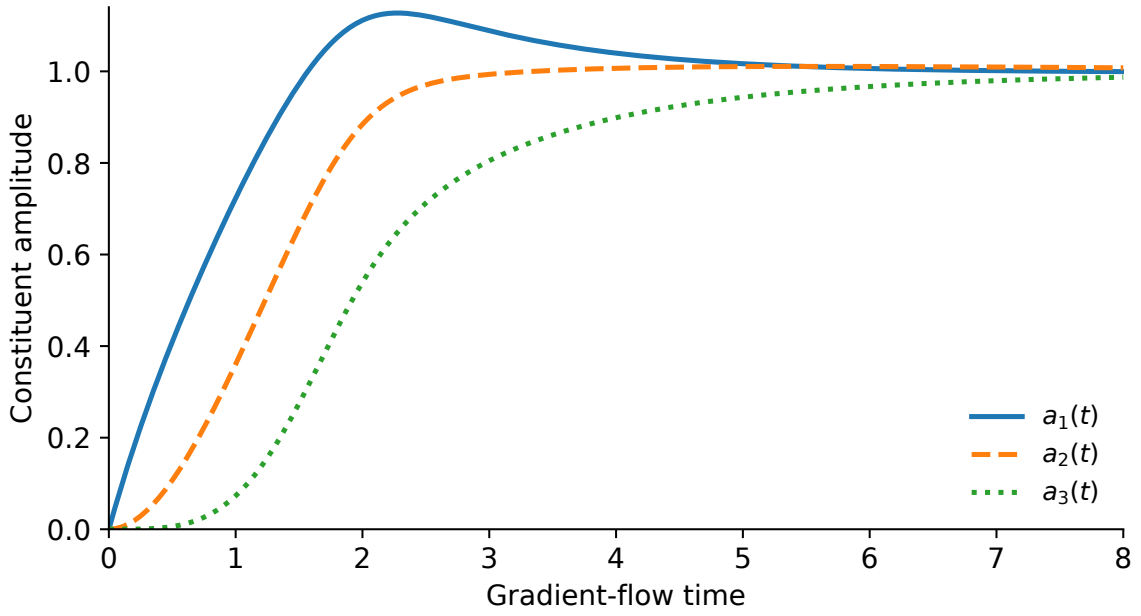


Figure 1: Joint staircase gradient flow, with no manual learning schedule. The early-time ordering follows from the theorem. A particular finite-time numerical trajectory does not establish universal ordering at all thresholds or weights.

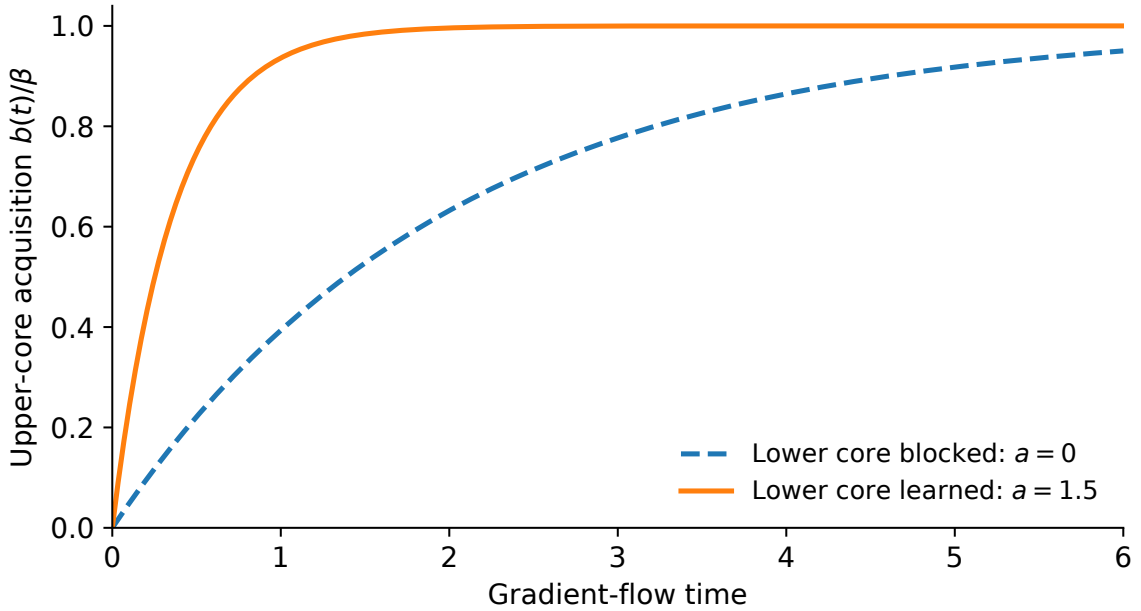


Figure 2: Soft upper-core acceleration in the exact three-leaf example. Both runs acquire the upper pairwise feature; the run with the lower core blocked still cannot represent the full triple interaction. The two curves are controlled interventions, not the natural joint trajectory.

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